

1 Explainability in Reinforcement Learning

2 Maxime Alaarabiou

3 Saturday 1st August, 2026

Chapter 1

Clustering Agent Representations

As discussed in the previous chapter, current explainability methods for Reinforcement Learning still lack intrinsic approaches capable of explaining the internal mechanisms of deep neural policies. To the best of our knowledge, only [Zahavy et al., 2017] have proposed exploiting the latent space learned by neural policies in order to identify the concepts used by the agent. Its main idea is to analyze the latent representations produced by the neural network and organize them into clusters that correspond to high-level concepts.

While the core idea of this paper is innovative and closely aligned with our vision of explainability in RL, this approach suffers from two important limitations. First, the clusters are manually constructed. Such a procedure heavily depends on the user’s expertise and neither guarantees reproducibility or scales. Second, no evaluation protocol is proposed to assess the quality of the obtained clusters. Their relevance is essentially determined through visual inspection.

The Clustering Agent Representations (CAR) method proposed in this chapter addresses these two limitations. It automates concept discovery by relying on unsupervised clustering algorithms to identify the concept structure of the latent space. It introduces an objective evaluation metric based on the stability of the extracted representations across independently trained policy.

The remainder of this chapter is organized as follows. The first section introduces the theoretical assumptions underlying the proposed CAR method, referred to as the representation convergence hypothesis. The second section describes the CAR method itself, with a detailed account of the extraction of latent representations using surrogate policies, along with the automatic detection of concept structures via clustering. The third section then presents the proposed evaluation metric.

1.1 Representation Convergence Hypothesis

The Clustering Agent Representations (CAR) method relies on a assumption, referred to as the Representation Convergence Hypothesis. This hypothesis follows the two assumptions established previously throughout this manuscript.

The first assumption rejects the notion that deep neural networks constitute irreducible black boxes ???. Adopting a mechanistic perspective, we consider that every process results from a sequence of causal computations. Although these computations may be highly complex, they remain explainable through the analysis of the internal mechanisms.

The second assumption states that neural networks achieve generalization by learning concepts ?. During optimization, the network progressively organizes its latent space into representations that capture the information required to solve the task.

Following these hypotheses, it becomes possible to explain the Neural Network (NN)’s process by understanding the organization of its latent space. This family of approaches is commonly referred to as mechanistic interpretability.

The Representation Convergence Hypothesis extends this mechanistic perspective by introducing an assumption regarding concepts that arise from learning. We hypothesize that, for a given task and a given neural network architecture, there exists a privileged conceptual organization. This organization naturally emerges because it constitutes an efficient representation for solving the task. Under this hypothesis, this means that independently trained neural networks converge toward the same concept structure.

It is important to clarify that this hypothesis does not imply convergence at the parameter level. Independently trained models are expected to converge toward different sets of weights. The proposed hypothesis concerns a more abstract level of representation. We assume that, although the numerical implementation of the networks may differ, the conceptual organization induced by their Latent Representation (LR) converges toward an equivalent conceptual structure. Consequently, the objective is not to compare neural parameters or latent activations directly, but compare the underlying conceptual structure encoded within the latent space.

The remainder of this chapter builds upon this hypothesis. This principle forms the theoretical foundation of the Clustering Universality metric introduced later.

1.2 Method

As discussed in Chapter ??, several hypotheses have been proposed regarding the organization of latent representations in deep neural networks. Following the work of [Zahavy et al., 2017], we therefore analyze the latent space through clustering techniques, placing our approach within the broader field of Deep Clustering (Section ??).

This design choice is motivated by our rejection of the superposition hypothesis and, more generally, of any view that assumes a linearly organized concept structure. We consider this premise unnecessarily restrictive. We only assume that concept representations induce a geometric structure that can be recovered through unsupervised clustering.

The CAR methodology is composed of two main stages. The first stage consists of generating multiple independent neural networks that implement the same decision policy, referred to as surrogate policies. The second stage clusters their latent representations in order to identify the underlying conceptual organization.

1.2.1 Multiple surrogate policies

The core idea of CAR is to evaluate whether independently trained neural networks converge toward similar conceptual organizations. To test this hypothesis, several neural networks must be trained independently and their latent representations compared. If similar conceptual structures consistently emerge, this provides empirical evidence supporting the Representation Convergence Hypothesis.

In supervised learning, this protocol is straightforward since several models can be independently trained on the same target function. Reinforcement learning, introduces an additional difficulty. Because of stochastic exploration, independently trained agents are not guaranteed to converge toward the same policy. Even if they achieve comparable performance, they may solve the task using different strategies. Making it impossible to determine whether differences observed in the latent space originate from the learning dynamics or from genuine differences between the learned policies.

To overcome this limitation, we do not directly compare independently trained reinforcement learning agents. Instead, we first consider a reference policy π that has already been trained. This policy is then imitated using several independently initialized surrogate neural networks trained through supervised behavioral cloning. Since all surrogate models approximate the same target function, they are expected to converge toward equivalent decision policies while preserving independent internal parameterizations.

This protocol provides an experimental setting in which differences between latent representations can be interpreted as resulting from the internal organization learned by each neural network rather than from differences in the underlying policy itself. It becomes possible to investigate whether equivalent policies also exhibit equivalent conceptual organizations.

It is important to emphasize that the use of surrogate policies does not make CAR a surrogate explainability method. In the explainability literature, surrogate approaches replace a complex model with a simpler and inherently interpretable model, such as a decision tree or a linear classifier. This is not the case here. The surrogate policies remain deep neural networks. Their role is not to simplify the decision process but to provide multiple independent realizations of the same policy function, enabling the study of representation convergence. The explanations are therefore still extracted directly from the internal representations of deep neural networks, making CAR a fully intrinsic and mechanistic explainability approach.

1.2.2 Training surrogate policies

Throughout the remainder of this manuscript, the initial policy we aim to explain is denoted as π (represented in green in Figure 1.1). This initial policy is obtained without imposing any constraints on its origin. It may result from classical or deep reinforcement learning [?], alpha-beta algorithm [?], or even from demonstrations provided by humans [Ross and Bagnell, 2010]. The crucial requirement is the availability of a dataset containing a sufficient number of projections from the observation space $\mathcal{O}$ to the action space $\mathcal{A}$ induced by π .

The replicated policies are referred to as surrogate policies, denoted by π' (shown in red in Figure 1.1). For any observation O , the distribution over the

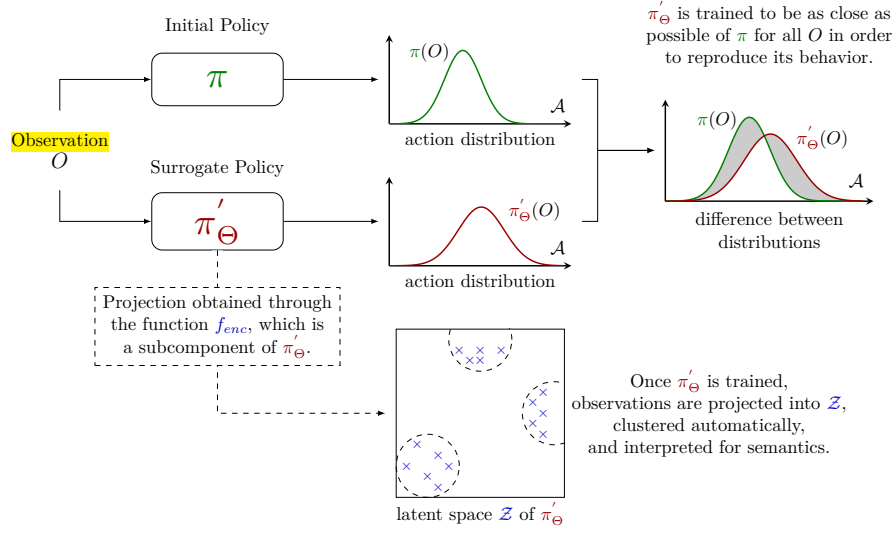


Figure 1.1: Illustration of the Clustering Agent Representation method.

131 action space produced by π' should approximate that of π as closely as possible.
 132 The surrogate policy π' is implemented as a NN parameterized by Θ . This
 133 network is composed of two principal components: an encoder and a decoder. The
 134 encoder, represented by the function f_{enc} , maps observations from $\mathcal{O}$ into a Latent
 135 Space (LS) $\mathcal{Z}$ (shown in blue in Figure 1.1). The decoder then reconstructs
 136 actions from this latent representation. Figure 1.2 details this architecture,
 137 showing how the encoder layers progressively compress the observation into the
 138 embedding before the decoder reconstructs the action distribution parameters
 139 from it.

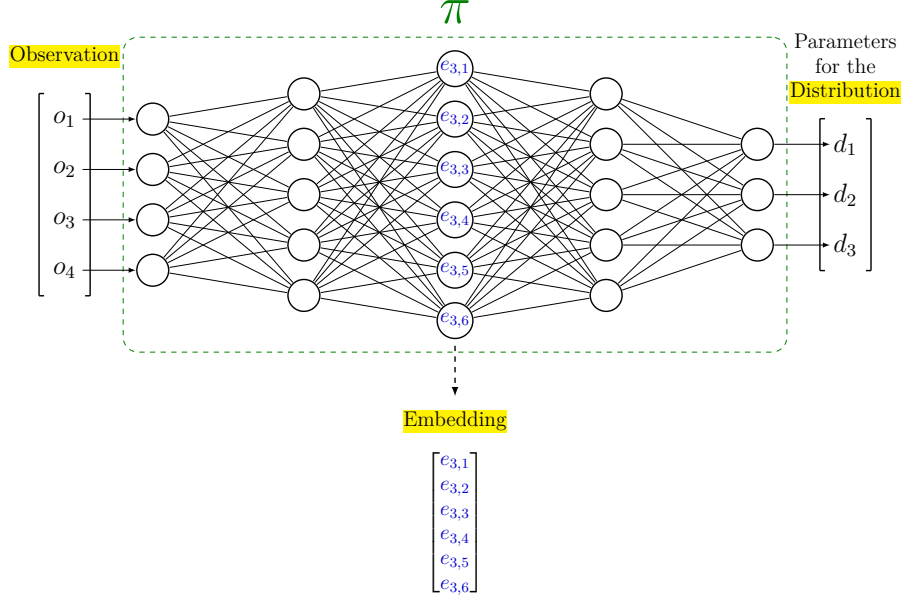


Figure 1.2: Architecture of the surrogate policy π : the encoder maps the observation O to the embedding Z , from which the decoder reconstructs the action distribution parameters.

140 To train π' to accurately replicate the action distribution, we define a global
 141 loss function $\mathcal{L}$ optimized via gradient descent. This loss, defined in Equa-
 142 tion (1.1), is computed as the Mean Squared Error (MSE) between the param-
 143 eters of the action distributions, obtained by applying the function p , which maps
 144 a distribution to its parameterization. In practice, this function does not need to
 145 be explicitly defined, as modern neural policies directly output the parameters
 146 of the action distribution. Once training is complete, this loss ensures that π'
 147 closely approximates the behavior of the original policy π .

$$\min_{\Theta} \sum_{O \in \mathcal{O}} \mathcal{L}(O) = \min_{\Theta} \sum_{O \in \mathcal{O}} \text{MSE}(p(\pi(O)), p(\pi'_{\Theta}(O))) \quad (1.1)$$

148 We denote by $\mathcal{E}$ the set of embeddings obtained by applying f_{enc} to a subset
 149 of observations $\mathbb{O} \subseteq \mathcal{O}$.

150 On the set $\mathcal{E}$, we use a clustering function f_{clust} to partition the point cloud
 151 of embedded observations into distinct clusters. Each observation is assigned
 152 a cluster label based on its proximity in the LS, resulting in the clustering $\mathcal{C}$.
 153 This process produces a structured representation of the groups of observations
 154 encountered by the agent, effectively simplifying the environment's representation
 155 through the lens of the policy's LS.

156 To demonstrate this clustering faithfully reflects the agent's internal decision-
 157 making process, we introduce a set of evaluation metrics, which will be detailed
 158 in the following section.

1.3 Evaluation Metrics

A key contribution of this work is the development of three evaluation metrics, designed to assess the effectiveness of CAR:

- **Behavior Reconstruction** : it measures how well a surrogate policy faithfully reproduces the behavior of the original policy by comparing their cumulative rewards.
- **Clustering Universality** : it evaluates the stability of clusterings by measuring the similarity of clusters obtained from independently trained surrogate policies.
- **Abstraction Score** : it quantifies the extent to which LS clusters capture behavioral abstractions rather than directly reflecting the structure of observations or actions.

The remainder of this section details the motivation for each metric and describes how they are calculated.

The following notations will be adopted for the rest of the manuscript: we consider n surrogate policies, denoted as $\pi'_1, \dots, \pi'_n$, trained as described in Section 1.2. Each surrogate policy is associated with a projection function $f_{enc_1}, \dots, f_{enc_n}$, which is applied to the set of observations $\mathbb{O}$, yielding embeddings $\mathcal{E}_1, \dots, \mathcal{E}_n$. A clustering function is then applied to each embedding set, producing partitions $\mathcal{C}_1, \dots, \mathcal{C}_n$.

1.3.1 Behavior Reconstruction

We argue that explaining a function b that closely approximates another function a is essentially the same as explaining a . In other words, if b accurately captures the key decision patterns of a , understanding b provides meaningful insights into how a works [Guidotti et al., 2018]. In our setting, this means that generating explanations for π' that faithfully reproduce π amounts to explaining π directly. To verify that π' is indeed a faithful copy, we introduce the *Behavior Reconstruction* metric.

This metric is computed by comparing the cumulative rewards obtained by the original policy π and the surrogate policies π' over m evaluation episodes. The behavior reconstruction metric is then defined as:

$$\text{Behavior Reconstruction} = \frac{1}{n \sum_{i=1}^m g_{\pi}^i} \sum_{j=1}^n \sum_{k=1}^m g_{\pi'_j}^k \quad (1.2)$$

In imitation learning, it is unlikely to expect the surrogate policy π' to outperform the initial policy π it aims to approximate through supervised learning [Ross and Bagnell, 2010]. Consequently this metric ranges in $[0, 1]$. Values approaching 1 indicate minimal loss in reproducing π , while values close to 0 indicate poor reproduction.

We deliberately adopt a reward-based definition rather than alternatives based on actions or action distributions (e.g., training loss, accuracy, KL divergence [Shlens, 2014], or optimal transport [Peyré and Cuturi, 2020]). Since surrogate

198 policies are trained in a supervised manner, they cannot develop strategies
 199 beyond those demonstrated by the initial policy. This means that if the rewards
 200 achieved are comparable, the surrogate policy is effectively copying the original
 201 policy’s behavior.

202 1.3.2 Clustering Universality

203 Our methodology involves analyzing the representations of an agent. For these
 204 representations to serve as a valid object of study in explainable artificial intel-
 205 ligence, they must demonstrate consistency across independent training runs.
 206 Indeed, if each surrogate π' produces substantially different representations,
 207 it is meaningless to search for recurring structures to interpret. Conversely,
 208 if independently trained surrogates yield consistent representations, the re-
 209 sulting structures are more likely to reflect intrinsic properties of the original
 210 policy rather than artifacts of training. This stability of discovered representa-
 211 tions across runs is referred to as **Universality** in mechanistic interpretability
 212 [Templeton et al., 2024].

213 In our study, agent representations take the form of clusters in the latent
 214 space. To assess the stability of these representations, we measure the similarity
 215 of clusters produced by different surrogate policies π' using the Adjusted Rand
 216 Index (ARI) [Hubert and Arabie, 1985]. For this reason, we refer to our metric
 217 as *Clustering Universality*, which evaluates the similarity between clusterings as
 218 follows:

$$\text{Clustering Universality} = \frac{1}{\binom{n}{2}} \sum_{i=1}^n \sum_{j=0}^{i-1} \text{ARI}(\mathcal{C}_i, \mathcal{C}_j) \quad (1.3)$$

219 This metric takes values in $[-1, 1]$. A value close to 1 indicates that surrogate
 220 policies produce highly similar partitions, suggesting that the clusters reflect
 221 intrinsic properties of the policy. Conversely, a value near 0 implies that the
 222 similarity between partitions is no better than that of two randomly generated
 223 clusterings, meaning that the clustering fails to capture any meaningful structure
 224 in the policy. A negative value, on the other hand, signals that the agreement
 225 between partitions is worse than what would be expected by chance, indicating
 226 a systematic disagreement between the clusterings.

227 For further details on the properties of the ARI, we refer the reader to [Hubert and Arabie, 1985,
 228 Warrens and van der Hoef, 2022] and to Appendix ???. Nevertheless, it is gener-
 229 ally accepted that an ARI score above 0.5 indicates a reasonably strong agreement
 230 between two clusterings [Amezquita et al., 2020].

231 1.3.3 Abstraction Score

232 If *Clustering Universality* is high, it becomes important to determine what
 233 underlies this stability. Two possibilities arise:

- 234 (i) the LS merely reproduces the structure of the observation space or action
 235 distribution, offering no abstraction;
- 236 (ii) the LS captures similarities in decision-making processes, yielding behav-
 237 iorally meaningful clusters.

238 To distinguish among these cases, we introduce the *Abstraction Score*. This
 239 metric contextualizes latent clusterings by examining how their patterns align
 240 with reference structures derived from the raw observation and action data. We
 241 consider : $\mathcal{C}_{obs}$, the clustering in the observation space $\mathbb{O}$, and $\mathcal{C}_{act}$, the clustering
 242 in the action-space distribution parameters. The clusterings $\mathcal{C}_{obs}$ and $\mathcal{C}_{act}$ are
 243 performed using the same algorithm, distance metric, and hyperparameters as
 244 those used for $\mathcal{C}_i$ in the LS, ensuring that any biases introduced by the clustering
 245 procedure itself are minimized.

246 For each latent clustering $\mathcal{C}_i$, we define its similarity to these baselines using
 247 the ARI:

$$\begin{aligned} A_i^{obs} &= \text{ARI}(\mathcal{C}_i, \mathcal{C}_{obs}) \\ A_i^{act} &= \text{ARI}(\mathcal{C}_i, \mathcal{C}_{act}) \end{aligned}$$

248 The *Abstraction Score* is subsequently calculated as one minus the average of
 249 the maximum absolute similarity values across these two reference baselines:

$$\text{Abstraction Score} = 1 - \frac{1}{n} \sum_{i=1}^n \max(|A_i^{obs}|, |A_i^{act}|) \quad (1.4)$$

250 The *Abstraction Score* ranges from $[0, 1]$. A score around 0 means that the
 251 latent clusters are strongly aligned with either the observation or the action
 252 space, reflecting minimal abstraction, cases (i). A score close to 1 indicates that
 253 the latent clusters are independent of both the observation and action spaces,
 254 capturing abstractions that meaningfully reflect the agent’s behavior, case (ii).

255 The following chapter presents the results of applying these metrics to specific
 256 examples.

Bibliography

- [Amezquita et al., 2020] Amezquita, R. A., Lun, A. T., Becht, E., Carey, V. J., Carpp, L. N., Geistlinger, L., Marini, F., Rue-Albrecht, K., Risso, D., Soneson, C., et al. (2020). Orchestrating single-cell analysis with Bioconductor, volume 17. Nature Publishing Group US New York. **Book**.
- [Guidotti et al., 2018] Guidotti, R., Monreale, A., Ruggieri, S., Turini, F., Pedreschi, D., and Giannotti, F. (2018). A survey of methods for explaining black box models. **Article**.
- [Hubert and Arabie, 1985] Hubert, L. and Arabie, P. (1985). Comparing partitions. Journal of classification, 2(1):193–218. **Article**.
- [Peyré and Cuturi, 2020] Peyré, G. and Cuturi, M. (2020). Computational optimal transport. **Article**.
- [Ross and Bagnell, 2010] Ross, S. and Bagnell, D. (2010). Efficient reductions for imitation learning. In Teh, Y. W. and Titterton, M., editors, Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics, volume 9 of Proceedings of Machine Learning Research, pages 661–668, Chia Laguna Resort, Sardinia, Italy. PMLR. **Article**.
- [Shlens, 2014] Shlens, J. (2014). Notes on kullback-leibler divergence and likelihood. **Article**.
- [Templeton et al., 2024] Templeton, A., Conerly, T., Marcus, J., Lindsey, J., Bricken, T., Chen, B., et al. (2024). Scaling monosemanticity: Extracting interpretable features from claude 3 sonnet. Transformer Circuits Thread. **Web**.
- [Warrens and van der Hoef, 2022] Warrens, M. J. and van der Hoef, H. (2022). Understanding the adjusted rand index and other partition comparison indices based on counting object pairs. Journal of Classification, 39(3):487–509. **Article**.
- [Zahavy et al., 2017] Zahavy, T., Zrihem, N. B., and Mannor, S. (2017). Graying the black box: Understanding dqns. **Article**.